

Zander W. Blasingame

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Professional Experience

AITHYRA <i>Vienna, Austria</i>	Postdoctoral Fellow <i>Supervisor: Alexander Tong</i> • Stochastic flow maps, likelihoods for continuous normalizing flows, AI for Science	2026 – Present
Clarkson University <i>Potsdam, NY</i>	Graduate Research Assistant <i>Supervisor: Chen Liu</i> • Diffusion models, neural differential equations, flow-based generative modeling • Face morphing attacks, biometrics, anomaly detection	2018 – 2025
Clarkson University <i>Potsdam, NY</i>	Undergraduate Research Assistant <i>Supervisor: Chen Liu</i> • GAN inversion, probability density estimation, zero-day malware detection	2016 – 2018
Griffiss Institute <i>Rome, NY</i>	Engineering Intern	05.2016 – 08.2016
UNH InterOperability Lab <i>Durham, NH</i>	Engineering Intern	07.2014 – 08.2014

Education

Clarkson University <i>Potsdam, NY</i>	Ph.D. in Electrical and Computer Engineering <i>Thesis: On Guided and Reversible Solvers for Neural Differential Equations</i>	2025
	M.Sc. in Electrical and Computer Engineering (en route to Ph.D.)	2025
	B.Sc. in Computer Engineering, <i>with Great Distinction</i>	2018

Publications

In reverse chronological order *Denotes equal contribution †Denotes equal advising

NeurIPS <i>Oral</i>	Strong Stochastic Flow Maps <i>S. McCallum*, Z.W. Blasingame*, T. Herschell, N. Rindtorff, A. Tong†, and J. Foster†</i> Published at NeurIPS (2026) · <i>Oral [top 0.4% of submissions]</i> <i>Presented at ICML 2026 SPIGM workshop</i> arxiv openreview code pdf slides video	2026
NeurIPS <i>Workshop</i>	Improved Training of Strong Stochastic Flow Maps <i>Z.W. Blasingame, A. Franca, N. Rindtorff, and A. Tong</i> <i>Presented at NeurIPS 2026 STODY workshop</i> openreview pdf	2026
NeurIPS <i>Workshop</i>	Global Transport Couplings for Classifier-Free Guided Flows <i>K. Petrović, Z.W. Blasingame, D. Rehman, İ.İ. Ceylan, M.M. Bronstein, S.Y. Zhang†, L. Atanackovic†, and A. Tong†</i> <i>Presented at NeurIPS 2026 STODY workshop</i> openreview pdf	2026

NeurIPS Workshop	Normalizing Flow Maps <i>FM. Ruscio[*], T.J. Hitge[*], Z. Peng, D. Rehman, Z.W. Blasingame, M.M. Bronstein, N.M. Boffi, A. Tong[†], and J. Bose[†]</i> <i>Presented at NeurIPS 2026 STODY workshop</i> openreview pdf	2026
ICML Oral	Rex: A Family of Reversible Exponential (Stochastic) Runge–Kutta Solvers <i>Z.W. Blasingame and C. Liu</i> Published at ICML (2026) · <i>Oral [top 0.7% of submissions]</i> <i>Presented at ICLR 2026 DeLTa and ReALM-GEN workshops</i> arxiv openreview code website pdf poster slides	2026
ICLR Workshop	Stochastic Few-step Models <i>R. Passaro, Z.W. Blasingame, M.M. Bronstein, and A. Tong</i> <i>Presented at ICLR 2026 DeLTa and ReALM-GEN workshops</i> openreview pdf	2026
NeurIPS	Greed is Good: A Unifying Perspective on Guided Generation <i>Z.W. Blasingame and C. Liu</i> Published at NeurIPS (2025) <i>Presented at ICML 2025 EXAIT workshop and ICLR 2025 FPI workshop</i> arxiv openreview pdf poster slides	2025
IEEE ASAP	Scalable Malware Detection Framework Using Performance Counters and Gradient Boosting <i>C. Woralert, C. Liu, and Z. Blasingame</i> Published at IEEE ASAP (2025)	2025
HASP	Feasibility of Scalable Performance Counter Malware Detection Framework Using LightGBM <i>C. Woralert, C. Liu, and Z. Blasingame</i> Published at HASP (2025)	2025
IEEE MOST	Poster: Adapting Pretrained Vision Transformers with LoRA Against Attack Vectors <i>R.E. Neddo, S. Willis, Z.W. Blasingame, and C. Liu</i> Published at IEEE MOST (2025)	2025
NeurIPS	AdjointDEIS: Efficient Gradients for Diffusion Models <i>Z.W. Blasingame and C. Liu</i> Published at NeurIPS (2024) <i>Presented at NeurIPS - New Frontiers in Adversarial Machine Learning</i> arxiv openreview code website pdf poster slides	2024
IEEE S&P	Fast-DiM: Towards Fast Diffusion Morphs <i>Z.W. Blasingame and C. Liu</i> Published at IEEE S&P (2024) arxiv code website pdf	2024
IJCB Spotlight	Greedy-DiM: Greedy Algorithms for Unreasonably Effective Face Morphs <i>Z.W. Blasingame and C. Liu</i> Published at IJCB (2024) · <i>Spotlight</i> arxiv code website pdf poster	2024

IEEE TBIOM	Leveraging Diffusion for Strong and High Quality Face Morphing Attacks <i>Z.W. Blasingame and C. Liu</i> Published at IEEE TBIOM (2024) <i>Presented at IJCB (Oral)</i> arxiv code website pdf poster slides	2024
IJCB	The Impact of Print-Scanning in Heterogeneous Morph Evaluation Scenarios <i>R.E. Neddo, Z.W. Blasingame, and C. Liu</i> Published at IJCB (2024) arxiv pdf poster	2024
HASP	Towards Effective Machine Learning Models for Ransomware Detection via Low-Level Hardware Information <i>C. Woralert, C. Liu, and Z. Blasingame</i> Published at HASP (2024)	2024
IEEE TCAS-I	HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware <i>C. Woralert, C. Liu, and Z. Blasingame</i> Published at IEEE TCAS-I (2023)	2023
AsianHOST <i>Oral</i>	A Comparison of One-class and Two-class Models for Ransomware Detection via Low-level Hardware Information <i>C. Woralert, C. Liu, Z. Blasingame, and Z. Yang</i> Published at AsianHOST (2023) · <i>Oral</i>	2023
AsianHOST <i>Best Paper Nominee</i>	HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware <i>C. Woralert, C. Liu, and Z. Blasingame</i> Published at AsianHOST (2022) · <i>Best Paper Nominee</i>	2022
IJCB <i>Oral</i>	Leveraging Adversarial Learning for the Detection of Morphing Attacks <i>Z. Blasingame and C. Liu</i> Published at IJCB (2021) · <i>Oral</i> slides	2021
ICANN <i>Oral</i>	Feature Creation Towards the Detection of Non-control-Flow Hijacking Attacks <i>Z. Blasingame, C. Liu, and X. Yao</i> Published at ICANN (2021) · <i>Oral</i> slides	2021
HASP	Detecting Non-Control-Flow Hijacking Attacks Using Contextual Execution Information <i>G. Torres, Z. Yang, Z. Blasingame, J. Bruska, and C. Liu</i> Published at HASP (2019)	2019
RESEC	Detecting Data Exploits Using Low-Level Hardware Information: A Short Time Series Approach <i>C. Liu, Z. Yang, Z. Blasingame, G. Torres, and J. Bruska</i> Published at RESEC (2018)	2018
SPIE	Verification of OpenSSL Version via Hardware Performance Counters <i>J. Bruska, Z. Blasingame, and C. Liu</i> Published at SPIE (2017)	2017

Invited Talks

AITHYRA <i>Vienna, Austria</i>	Strong Stochastic Flow Maps <i>AI x Bio Seminar. Organized by Michael Bronstein and Georg Winter</i> slides	07.2026
MolSS <i>Online</i>	Strong Stochastic Flow Maps <i>Machine Learning for Molecule Simulations & Sampler Reading Group</i> slides	07.2026
Starkly Speaking <i>Online</i>	Strong Stochastic Flow Maps <i>Starkly Speaking Reading Group. Organized by Hannes Stärk</i> slides video	07.2026
Mila <i>Montréal, Canada</i>	AdjointDEIS: Efficient Gradients for Diffusion Models <i>Sampling Reading Group. Organized by Kirill Neklyudov</i> slides	01.2025
Embassy of the Kingdom of the Netherlands <i>Washington, D.C.</i>	Diffusion Morphs (DiM): Diffusion is all you need for highly effective face morphs <i>Transatlantic Dialogue on Presentation Attack Detection</i> slides	11.2024
IEEE IJCB <i>Buffalo, NY</i>	Diffusion Morphs (DiM): Leveraging Diffusion for Strong and High-Quality Face Morphing Attacks <i>Journal Presentation</i> slides	09.2024
Idiap <i>Martigny, Switzerland</i>	Diffusion Morphs (DiM): The Power of Iterative Generative Models for Attacking FR Systems <i>Biometrics Security & Privacy Group. Organized by Sébastien Marcel</i> slides	07.2024
DSA <i>Online</i>	Diffusion for the Generation of Face Morphs <i>CITeR & Document Security Alliance</i> slides	02.2024

Awards and Honors

NeurIPS Oral Presentation [top 0.4% of submissions]	2026
ICML Oral Presentation [top 0.7% of submissions]	2026
ICML Gold Reviewer Award	2026
NeurIPS Top Reviewer Award	2025
AISTATS Best Reviewer Award	2025
IJCB Doctoral Consortium	2024
Best Oral Presentation for Software Engineering — Clarkson RAPS	2017
Clarkson Presidential Scholar	2015 – 2018

Teaching

Griffiss Institute <i>Rome, NY</i>	Instructor • Cyber Summer Camp • Advanced Cyber Summer Camp	2020 – 2024
Clarkson University <i>Potsdam, NY</i>	Graduate Teaching Assistant • Electrical Science • Digital Design • ECE Sophomore Lab	2018 – 2021
Clarkson University <i>Potsdam, NY</i>	Undergraduate Teaching Assistant • Software System Architecture • Differential Equations	2016 – 2018

Funding

CITeR	Explainable Image Quality with Transformer-based Models <i>NSF Industry–University Cooperative Research Center. Primary student investigator. PI: Chen Liu</i>	2023 – 2024
CITeR	Towards the Creation of a Large Dataset of High-Quality Face Morphs <i>NSF IUCRC. Primary student investigator at Clarkson. PIs: Chen Liu, Stephanie Schuckers, Xin Li, Jeremy Dawson, Nasser Nasrabadi, David Doermann, Srirangaraj Setlur, Siwei Lyu, Xiaoming Liu, Sébastien Marcel</i>	2021 – 2024
CITeR	Comparative Detection of Facial Image Manipulation Techniques <i>NSF IUCRC. Primary student investigator. PI: Chen Liu</i>	2020 – 2022
CITeR	Adversarial Learning Based Approach Against Face Morphing Attacks <i>NSF IUCRC. Primary student investigator. PI: Chen Liu</i>	2019 – 2021

Professional Services

Area Chair	NeurIPS Workshop on AI for Stochastic Dynamics (2026)
Reviewer at NeurIPS	2024, 2025 (Top Reviewer), 2026
Reviewer at ICML	2025, 2026 (Gold Reviewer)
Reviewer at ICLR	2026
Reviewer at AISTATS	2025 (Best Reviewer)
Workshop Reviewer	NeurIPS Frontiers in Probabilistic Inference (2025); ICCV 2C000L (2025); ICLR Deep Generative Models (2025); ICLR Frontiers in Probabilistic Inference (2025); IJCB Face Morphing Attack and Detection Techniques (2024)
Other Reviewing	IEEE ICPR (2024); IEEE MOST (2023–2024); IEEE BTAS (2019)

Skills

ML Architectures	Diffusion Models, Flow Matching, Neural Differential Equations, GANs, VAEs
Distributed Computing	Slurm, CUDA, ROCm, Linux

Languages	Python, C/C++, Java, Javascript, Bash, MATLAB, VHDL
Frameworks	TensorFlow, Pytorch, Numpy, Scipy, Matplotlib, Plotly, Pandas, Jax
Tools	Vim, Git, Docker, LaTeX
Audio	Over a decade of experience as a FOH engineer including theater productions and worship services

References

References available upon request